



GUIDELINES

CAPSTONE PROJECT IN DATA SCIENCE II

Department of Data Science

Faculty of Computing

Sabaragamuwa University of Sri Lanka

2025/2026

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1. Introduction

1.1. Overview

The Capstone Project in Data Science II (DS3206) is a compulsory course module for the Bachelor of Science Honours Degree Programme in Data Science, offered by the Department of Data Science (DDS), Faculty of Computing, Sabaragamuwa University of Sri Lanka. This module is carried out individually by students during the second semester of the second year. It builds upon the foundational experience gained in Capstone Project I and focuses on developing advanced analytical, modeling, and visualization capabilities in solving real-world data science problems.

The project is designed to enable students to engage in the complete data science lifecycle, including problem formulation, data acquisition, data preparation, modeling, evaluation, and communication of results. Students will apply a broad range of analytical tools and techniques, including machine learning, statistical modeling, and data visualization, to derive insights from real or realistic datasets. Emphasis is placed on understanding domain-specific problems, designing data-driven solutions, and presenting findings effectively to technical and non-technical audiences.

Students are expected to clearly define their project goals, identify and justify the use of suitable datasets and methods, and demonstrate a deep understanding of the analytical process. They will also rehearse and refine their presentation skills in preparation for the final programme-level evaluation.

Each student is assigned an internal supervisor who will provide continuous guidance and feedback throughout the project. Students must maintain regular communication with their supervisor, track progress against key milestones, and submit a progress report detailing the status of their project. The final project outcomes will be assessed through a formal presentation and a comprehensive final report at the end of the semester.

The Capstone Project in Data Science II is equivalent to 2 credits, spans 15 weeks, and is evaluated by internal supervisors and a panel of evaluators. It is a critical component of the curriculum designed to synthesize and apply the theoretical and practical knowledge accumulated during the first two years of the degree programme.

1.2. Vision

To empower students to become innovative and ethical data science professionals by equipping them with the skills and mindset necessary to solve complex real-world problems through data-driven decision-making and impactful communication.

1.3. Mission

The mission of the Capstone Project in Data Science II is to cultivate independent, research-oriented thinking by guiding students through the complete data science lifecycle—from problem formulation to the communication of insights. This course empowers students to apply advanced analytical, statistical, and machine learning techniques to real-world datasets, fostering both technical expertise and practical problem-solving skills. It emphasizes the importance of ethical data practices, domain-specific understanding, and effective communication to diverse audiences. Through continuous mentorship and hands-on experience, the project aims to bridge academic learning with real-world application, preparing students to become competent and impactful data science professionals.

2. Objectives

Objectives of the Capstone Project in Data Science I:

- Apply advanced data analysis, statistical modeling, and machine learning techniques to real-world or domain-specific datasets.
- Deepen students' ability to extract meaningful insights from complex data through appropriate preprocessing, modeling, and visualization techniques.
- Encourage critical thinking and problem-solving by designing and evaluating data-driven solutions to real-life challenges.
- Enhance students' capability to communicate analytical findings effectively to both technical and non-technical stakeholders using data storytelling and visualization tools.
- Foster independent project management by tracking progress, setting milestones, and incorporating continuous feedback from supervisors.
- Strengthen the ability to reflect on and justify methodological choices, ethical considerations, and limitations of data-driven solutions.
- Prepare students for professional practice or postgraduate research by simulating real-world analytics workflows and emphasizing evidence-based decision-making.

3. Proposed Action Plan

Students are expected to undertake the Capstone Project in Data Science II (DS3206) during the second year of the degree programme. Although originally scheduled for Semester II, the project has been initiated earlier in Semester I due to time constraints and academic planning requirements. Prior to commencing the project, students must complete all prerequisite coursework at the beginning of Semester I. Once these requirements are fulfilled, students may proceed with the project planning and implementation phases under the supervision of an assigned academic mentor. Throughout the academic year, students are expected to make steady progress by adhering to defined milestones, regularly communicating with their supervisors, and incorporating feedback. The project will culminate at the end of Semester II, at which point students are required to submit all final deliverables, including the final report, and a formal presentation.

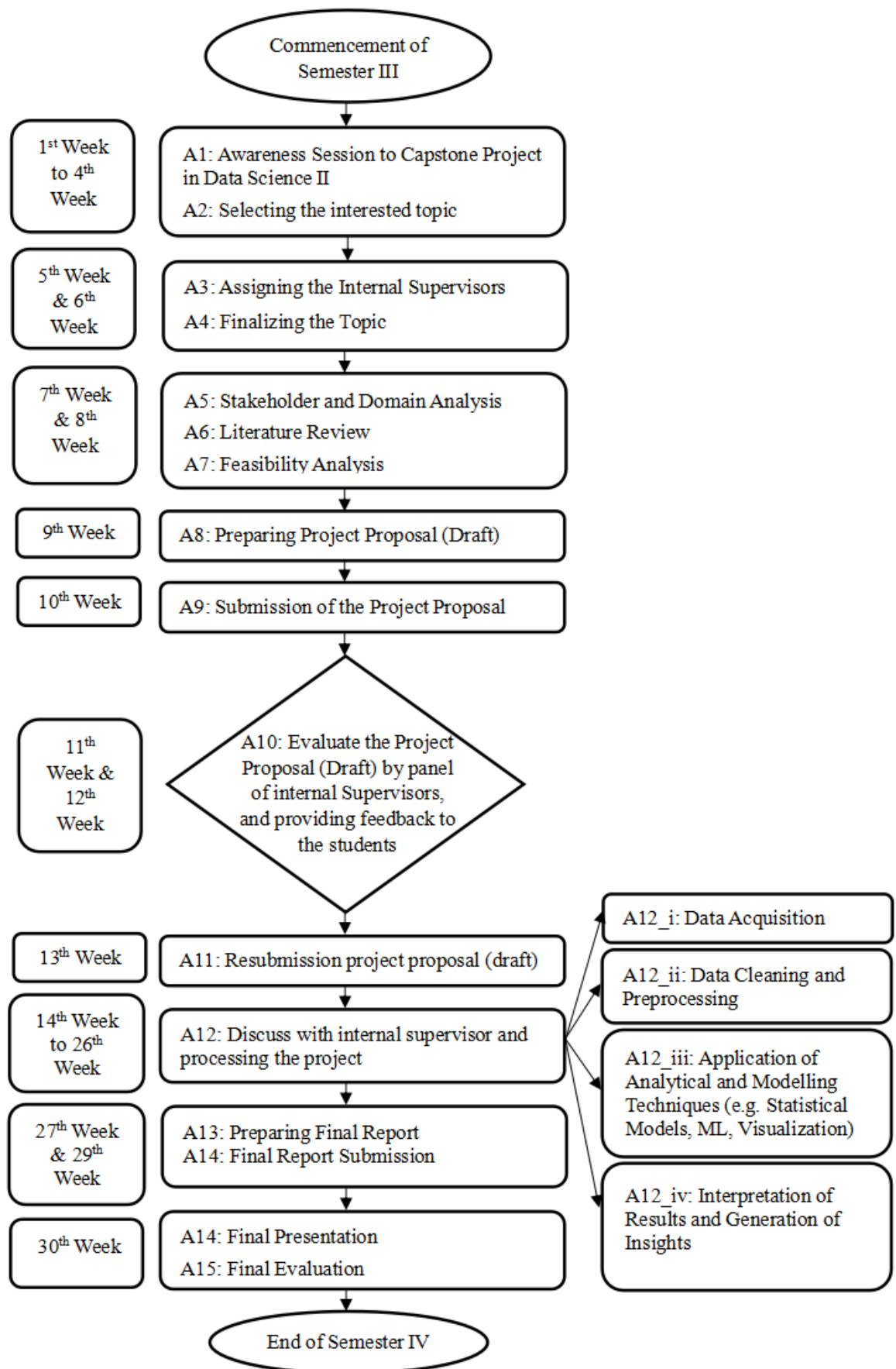


Figure 1: Proposed Action Plan

3.1. A1: Awareness Session to Capstone Project in Data Science II

An awareness session will be conducted to introduce students to the Capstone Project in Data Science II. This session will provide comprehensive guidance on the project's objectives, scope, expectations, evaluation criteria, and key timelines. Emphasis will be placed on the application of advanced analytical and modeling techniques, as well as the importance of domain relevance and impactful communication of findings.

3.2. A2: Selecting the Interested Topic

Students will be guided in selecting a new, relevant, and analytically rich real-world problem for their Capstone Project in Data Science II. They are encouraged to choose a topic that not only aligns with their academic and career interests but also presents opportunities to apply advanced data analysis, modeling, and visualization techniques. The selected topic should be significant, data-driven, and feasible within the given timeframe, allowing students to demonstrate both technical competence and critical thinking in solving complex problems.

3.3. A3: Assigning the Internal Supervisors

The department will assign an Internal Supervisor to each student based on their chosen project topics. Students should be aware of their assigned Internal Supervisor and are required to initiate contact with them to discuss their Capstone Project. This interaction is crucial for receiving guidance, feedback, and support throughout the project. The Internal Supervisor will assist students in refining their project approach, ensuring they stay on track, and addressing any issues that may arise during the course of the project.

3.4. A4: Finalizing the Topic

Students should finalize their project topic after conducting a feasibility analysis. This analysis involves evaluating the practicality and scope of the chosen topic, ensuring that it aligns with available resources, data accessibility, and project objectives. The finalized topic should be clearly defined and feasible for completion within the project's timeframe.

3.5. A5: Stakeholder and Domain Analysis

In this stage, students are required to conduct a comprehensive analysis of the domain context and identify key stakeholders relevant to their selected project topic. This involves understanding the roles, expectations, and influence of individuals or entities who are impacted by or have an interest in the project outcomes. Students are expected to assess how stakeholder needs can shape the data requirements, analytical approach, and interpretation of results, ensuring that the final solution remains relevant, ethical, and practically applicable.

3.6. A6: Literature Review

At this stage, students will conduct a thorough literature review to explore existing research studies, analytical models, methodologies, and tools related to their selected topic. The objective is to understand the current state of knowledge, identify gaps in existing solutions, and ensure that their project offers novelty or meaningful improvement in approach, application, or interpretation.

3.7. A7: Feasibility Analysis

In this phase, students will evaluate the practical feasibility of executing the proposed project within the given timeframe and available resources. This includes assessing the availability and quality of data, the suitability of selected tools and technologies, required computational resources, and time constraints.

3.8. A8: Preparing Project Proposal (Draft)

Students are required to prepare a draft project proposal in consultation with their assigned internal supervisor. The proposal must clearly define the objectives, scope, problem statement, methodology, tools, and expected outcomes of the proposed Capstone Project. The document must be structured and formatted in accordance with the prescribed departmental guidelines.

3.9. A9: Submission of the project Proposal

Students are required to submit a comprehensive proposal document that includes the project title, introduction, objectives, scope, methodology, and a tentative project plan. Students must submit their proposals by the given deadline.

3.10. A10: Evaluate the Project Proposal (Draft) by panel of Internal Supervisors, and providing feedback to the students

The proposal will undergo an initial review process, and feedback will be provided to guide any necessary revisions. This ensures that the project's scope, objectives, and methodology are clearly defined and approved before the project begins. Every student will be given suggestions and feedback about the proposal.

3.11. A11: Resubmission Project Proposal (Draft)

Students are responsible for resubmitting the project proposal by addressing all the comments and feedback from the reviewers.

3.12. A12: Discussion with Internal Supervisors and processing the Project

Following the incorporation of feedback from the proposal review panel, students will proceed to the implementation phase of their Capstone Project. Regular and structured discussions with the internal supervisor are essential at this stage to address any technical, analytical, or domain-specific challenges that may arise. The supervisor will provide continuous guidance, ensure that the project remains aligned with its objectives, and support students in applying appropriate data modeling, analysis, and visualization techniques to achieve meaningful results.

3.12.1. A12_i: Data Acquisition

This step involves identifying, sourcing, and collecting relevant datasets required for the project. Students may gather data from publicly available sources, institutional databases, APIs, or through web scraping—ensuring that the data aligns with the defined problem and is ethically obtained.

3.12.2. A12_ii: Data Cleaning and Preprocessing

Once the data is acquired, students must clean and preprocess it to make it suitable for analysis. This includes handling missing values, correcting inconsistencies, encoding categorical variables, normalization, and feature selection or extraction—ensuring the data is accurate, consistent, and analysis-ready.

3.12.3. A12_iii: Application of Analytical and Modelling Techniques (e.g. Statistical Models, ML, Visualization)

At this stage, students will apply appropriate statistical models, machine learning algorithms, and data visualization tools to analyze the data. The choice of techniques should align with the project goals and data type, and must be justified based on the literature and feasibility analysis.

3.12.4. A12_iv: Interpretation of Results and Generation of Insights

Students will interpret the output of their models to extract meaningful insights related to the problem statement. This includes evaluating model performance, understanding patterns or trends in the data, and relating results back to the stakeholder needs and domain context for informed decision-making.

3.13. A13: Preparing Final Report

Students are required to compile a comprehensive final report that documents all aspects of the Capstone Project. This includes the problem definition, literature review, methodology, data processing steps, analysis, results, and conclusions. The report should reflect critical thinking, technical rigor, and clarity, and must be formatted according to the prescribed departmental guidelines for submission and evaluation.

3.14. A14: Final Report Submission

Students must submit the finalized version of their Capstone Project report by the stipulated deadline. The report should incorporate all feedback received during the project process and must clearly present the problem addressed, data analysis workflow, methodologies applied, results obtained, and key insights. Proper citation, formatting, and adherence to academic writing standards are essential, as the final report serves as a key component of the project's overall evaluation.

3.15. A15: Final Presentation

The presentation should provide a concise yet detailed summary of the students' overall performance during the Capstone Project. It should be presented in an engaging and logically organized manner, effectively communicating the key aspects of the project. The presentation must be submitted at the beginning of the 30th week.

3.16. A16: Final Evaluation

The final evaluation of Capstone Project in Data Science II will be conducted by a panel of experienced evaluators familiar with the project's objectives and progression. This panel will assess students' overall performance, including their adherence to project guidelines, technical competence, innovation, and the quality of final deliverables such as the report and presentation. The evaluation will comprehensively consider both the project development process and the final outcomes, ensuring a fair and thorough assessment of the student's mastery of advanced data science skills.

4. Instructions for the Students and Supervisors

4.1. Instructions for Students

- This is an individual project which every student should complete.
- Students have to select the interested topic of the project during the 1st four weeks of the first semester of the second year.
- Students are allowed to select any interested project with trending/timely technologies.
- The Department will assign an Internal Supervisor to each student.
- If students encounter any issues, they must seek guidance from their assigned Internal Supervisor.
- Consequently, students should do a stakeholder analysis, literature review and feasibility analysis.
- Students must submit a project proposal, including project title, introduction, objectives, scope, methodology, and a tentative project plan.
- All the Project proposals will be reviewed by a panel assigned by the Department and students should address all the feedback given by the review panel students should resubmit the proposal within 7 days of the project proposal being reviewed.
- The proposal must be approved by the Department before the project can commence.
- Students must report to their internal supervisor once a week to show/update the progress of their work. If the students are not properly communicating with the internal supervisors, they have the power to take strict actions and report to the respective departments.
- Students should contact their internal supervisors at the first possible instance and waiting for the last moment will cause for late submissions and marks reductions.
- Students must submit their final project report along with the final outcomes, and final presentation to the final evaluation.
- Students should strictly follow the Guidelines provided by the Department to prepare each and every report.
- Students are encouraged to use Version Control Tools like GitHub and Project management tools like Trello, and Microsoft Projects.
- Students are strictly advice to work for the given deadlines. If anyone does not follow the rules, the department has the power to take strict actions and reduce the considerable number of marks for each submission/milestone. Departments will not be extending the deadlines for any reason.

4.2. Instructions for Internal Supervisors

- Every Student will get one Internal Supervisor to get guidance during their Capstone Project II.
- Internal supervisors may get more than one students to guide during the semester.
- Supervisors are expected to find out the solutions for their internal matters and they should directly involve evaluating the progress of the students.
- Internal Supervisors can schedule the meeting if needed and they can provide feedback on the progress of the project.
- The department will provide guidelines to prepare proposals, interim reports, final dissertations, and presentations. Internal Supervisors also have access to those resources and are expected to adhere to these guidelines during supervision.
- Internal Supervisors should participate in each and every evaluation throughout the project.
- Guidelines for Supervision:
 - Please note that all guidelines for the preparation of documents and presentations are provided to students. Therefore, please advise the students that they should adhere to those guidelines.
 - Please note that those guidelines use very generic terms such as Others' work, Approach, Design, etc. and you are required to guide students to find suitable terms for these as per their own projects.
 - The internal Supervisor should be aware of the students with the project objectives, process, and evaluation.
 - Please advise students about the importance of testing or evaluation in all scientific projects. You may guide the students to organize the evaluation in terms of evaluation strategy, identification of participants, controlling the experiments, design of questionnaires, and reporting the results. Please make sure that the evaluation has been structured to measure the objectives.
 - Always encourage students to work before the deadlines. If any students are not following the rules, internal supervisors have the power to take immediate strict actions and report to the project coordinator of department heads.

5. Guidelines

5.1. Capstone Project in Data Science II Proposal Guidelines

The project proposal for Capstone Project in Data Science II (DS3206) is a vital planning document that sets the direction for executing the data science project effectively. It must reflect a clear understanding of the problem domain, data-driven objectives, and the analytical methodologies to be used. The proposal should be developed in close consultation with the Internal Supervisor and must demonstrate both technical feasibility and domain relevance, ensuring the project can be completed within the stipulated timeframe.

The document must follow the structure below and maintain a clear, formal presentation style:

- Title of the Project
- Approval of Capstone Project II
- Index
- Acknowledgment
- Student Details
- Introduction & Objective of the Capstone Project II
- Problem Statement & Motivation
- Literature Review
- Stakeholder & Domain Analysis
- Feasibility Analysis (Time, Tools, Data Availability)
- Methodology (including data acquisition, modeling techniques, tools, and frameworks)
- Functional and Non-Functional Requirements
- Analytical Design (e.g., Workflow Diagrams, Data Flow, Data Models, Pipeline Architecture)
- Data Sources and Data Characteristics
- Project Plan and Timeline (including Gantt Chart and Milestones)
- Expected Outcomes
- Limitations and Risk Factors
- References

5.1.1. Guidelines for the Proposal Formatting

- Font face - Times New Roman.
- Font size- 11pt
- Line spacing - 1.5 line space

5.1.2. Template for the Approval sheet**Approval of Capstone Project in Data Science II**

1. Title of the Capstone Project:
2. Details of the Student:

Index No	Name with Initials	Email	Mobile No	Signature of Student

3. Name of the Internal Supervisor:
4. Internal Supervisor's designation:

 For office use only:

Approved/Not approved
 Signature of the Internal Supervisor
 Date:
 Suggest if any:

5.2. Capstone Project in Data Science II Final Report Guidelines

The final report is the most critical component of the Capstone Project in Data Science II (DS3206). It should be a comprehensive document with double spacing. Students are encouraged to use any suitable text editor to compose their project report. To ensure a high-quality final report, Capstone Project II Report should strictly follow the points given below:

5.2.1. Report Formatting

The information given in this section explains the formatting of the report. Use any word processing tools to set them up.

Layout

- Use A4 size with 35mm left margin 30mm right, top, and bottom margins
- Page numbers should be included (See Tables of Content)

Fonts

- Font type: Times New Roman
- Font Size: 12pt

Alignments

- Allow 1.15mm line-spacing
- Paragraphs should be separated by one blank line
- Permitted font color - only black
- Left and right margins should be justified

Legends

- Should be emboldened and centered

Page limit (Quantity can't substitute the Quality)

- There are not any limitations/strict rules about overall page count.
- Get advice from your respective supervisor regarding the page limit

Print

- The whole report should be printed on one side
- Type: Grayscale

Structure

- Follow the instructions given in Anatomy of the Report

5.2.2. Anatomy of the Report

Title Page

Generally, the title page contains a precise title, names, and affiliations of investigators along with institutional details. Don't do fancy though you are free to follow your own set of ideas.

Declaration

Type the particular declaration given in Appendix A.

Certificate of Approval

Appendix B

Acknowledgment

Even though it is optional, gratitude towards the people who gave the real contribution and support can be praised here (typically your supervisor, head of the department, friends and colleagues, etc.).

Abstract

The abstract is the shortened form or the summary of the complete report. Give more emphasis on results, recommendations, and conclusion and it should be more concise. Apart from these major points, the purpose of the report can also be included.

This section;

- can contain 250 - 300 words
- should be written as a single paragraph using the present tense
- Insert the keywords (4 to 6 relevant keywords) immediately after the abstract section.

Table of Contents

The arrangement of the complete report can be illustrated in the table of contents. Use Roman numbering except for primary chapters, for which the numerals can be used.

List of Figures

The list of figures with numbers and titles can be shown together with page numbers where the figures are located. Keep in mind that all figures and tables listed here must be referred to inside the main text.

E.g. "The use case diagram shown in Figure 1"

List of Tables

The list of all tables should be presented along with their corresponding page numbers. Tables must be organized by chapters to maintain a clear structure and enhance navigability.

Chapter 1: Introduction

This chapter provides an overall introduction to the Capstone Project in Data Science II. It connects all major sections of the report with a concise overview. The chapter should engage the reader and provide a strong rationale for the chosen project.

Subsections may include:

- Major goals and objectives of the project
- Motivation behind selecting the topic
- Defined scope of the project
- Chosen methodology or approach
- Brief summary of results and insights

The length of this chapter depends on the complexity and breadth of the chosen topic.

Chapter 2: Background and Problem Domain

This chapter sets the context by outlining the real-world problem, the relevance of the selected domain, and the need for a data-driven approach. It should also address gaps in current solutions and briefly discuss related work.

Subsections may include:

- Description of the domain or industry context
- Justification for the project topic
- Review of existing solutions and limitations

Chapter 3: Data Acquisition and Preparation

This chapter discusses how data was collected, cleaned, transformed, and made suitable for analysis.

Include details on:

- Data sources and acquisition methods
- Data cleaning and preprocessing steps
- Feature selection or extraction methods
- Summary statistics or visual overview of datasets

Chapter 4: Modeling and Analysis

This chapter explains how analytical and machine learning models were used to explore or solve the defined problem. Models must be described, justified, and evaluated.

Content may include:

- Description of statistical or machine learning models used
- Evaluation metrics and validation techniques
- Interpretation of results and performance
- Visualizations (e.g., plots, charts, confusion matrices)

Chapter 5: Results, Insights, and Discussion

This chapter presents findings obtained from the applied models and explains their relevance to the domain problem. It also discusses the insights generated, their practical implications, and any limitations observed.

Subsections may include:

- Summary of key results
- Interpretation and insights
- Limitations and potential biases
- Comparison with existing benchmarks or solutions

Chapter 6: Future Enhancements

Future directions and improvements for the project are discussed here. These suggestions can assist future students or researchers interested in building upon your work.

Include:

- Identified challenges and limitations
- Proposed technical enhancements or data extensions
- Areas for future exploration

Chapter 7: Conclusion

Summarize the work done, methods used, results achieved, and their significance. This chapter ties the entire report together and highlights the contributions of the project.

Include:

- Recap of goals and whether they were met
- Key findings and contributions
- Broader implications of the work

References

Provide a complete list of all references cited in the text using IEEE referencing style. Include books, research papers, articles, websites, and datasets. All cited materials must appear in this section.

Glossary

A list of special technical words or acronyms may be necessary. This is particularly true if the subject deals with a new area with a specialized vocabulary that the average reader in the discipline might not be familiar with, such as the Biotechnology. This list should come after the appendices.

Appendices (if applicable)

Appendices are supplementary documents to the main text. All appendices should be arranged in order, titled and alphabetical numbering can be used.

E.g. Appendix A, Appendix B and so on

- Appendices may include;
- Pieces of bulky research work/summaries of results obtained elsewhere
- Complete or partial data as tables
- Program listing
- Detailed maps, charts and diagrams

Appendix A - Sample Declaration page Declaration (18pt + B)

We declare that this thesis does not incorporate, without acknowledgment, any material previously submitted for a Degree or a Diploma in any University, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person or ourself except where due reference is made in the text. Also, we hereby grant to Sabaragamuwa University of Sri Lanka the non- exclusive right to reproduce and distribute my thesis, in whole or in part in print, electronic or other medium. We retain the right to use this content in whole or in part in future works (such as articles or books).

Index Number	Name of Student	Date	Signature of Student

Appendix B – Certificate of Approval

Certificate of Approval

We hereby declare that this thesis is from the student's own work and effort, and all other sources of information used have been acknowledged. This thesis has been submitted with our approval.

Index Number	Name of Student	Date	Signature of Student

Name of Internal Supervisor:

.....

Date:

Signature of Internal Supervisor

Name of Head of the Department:

.....

Date:

Signature of Head of the Department

6. Evaluation Procedure

To ensure proper conduction of each project, the progress of each project should be monitored on a continuous basis; by the internal supervisor, PDC and the evaluation panel. In order to do so, it is planned to hold three evaluations to be made by each project student in each semester. Entire evaluation will be taken by the same panel of evaluators from the beginning for each and every student.

6.1. Project Proposal Evaluation

This will be a proposal report, which will be reviewed by a panel of internal supervisors. In this report, students are required to provide a detailed document outlining the main aim/objective of the project, the methodology to be employed, the tentative work plan, and any supporting materials such as diagrams or charts.

If the report does not meet the required standards, the panel will ask the students, in consultation with their internal supervisor, to revise and resubmit the proposal within a week. Alternatively, if the committee deems the project insufficient in standard or feasibility, students may be required to select a new project topic. The internal supervisor will evaluate each student based on the quality of their proposal report, following the established project rubrics. The evaluation of the report will carry a weight of 30% of the total marks for the proposal.

6.2. Final Outcomes Demonstration to get the Internal Supervisors' recommendation & feedback for the final submission

Students must complete 100% of their project and demonstrate it to their respective internal supervisors to get approval for the final submission. The respective internal supervisors should check whether the final outcomes meet the given scope and will give feedback and recommendations for the final submission.

6.3. Final Report and Presentation Evaluation

The final evaluation for Capstone Project in Data Science II will be coordinated by the Project Development Committee (PDC) at the end of the project semester. The evaluation panel will consist of internal supervisors and representatives from the Department of Data Science who have been actively involved in monitoring the students' progress throughout the project period.

Each student is required to submit a comprehensive final report documenting the progress and key milestones achieved over the course of the 30-week project period. The report should concisely present the overall methodology, analytical and modeling techniques used, key findings, and insights derived from the project. It should also reflect on challenges faced and how they were addressed. This final report must be reviewed and signed by the assigned internal supervisor and submitted to the PDC at least two days before the final presentation (scheduled during the 30th week of the second semester).

Students are also required to deliver a final presentation before the evaluation panel. The presentation should include a complete walkthrough of the project objectives, data science process (data acquisition, cleaning, modeling, evaluation), results, insights, and overall conclusions. Students must also demonstrate the working solution or output they developed (e.g., model, dashboard, system, etc.).

In this presentation the panel is supposed to mark each student based on their project content, project outcome and performances, presentation made, project progress, Depth and effectiveness of data analysis and modeling, maintaining professionalism, and attendance. The final evaluation will account for 70% of the total marks and all evaluations will be conducted in accordance with the standardized Project Evaluation Rubrics.

CAPSTONE PROJECT IN DATA SCIENCE II PROPOSAL EVALUATION RUBRICS

LEVELS OF ACHIEVEMENTS (Total 100 marks)					
a.	Problem Statement & Motivation 10 Marks	Problem statement is clear, well defined, and addresses a significant issue. Motivation is strong, demonstrating deep understanding of the problem's impact and relevance. (10)	Problem statement is clear and relevant, though it may lack some depth. Motivation is good, but the significance of the problem could be better articulated. (08)	Problem statement is somewhat unclear or lacks detail. Motivation is present but weak, with limited discussion of the problem's impact or relevance. (05)	Problem statement is unclear, poorly defined, or irrelevant. Motivation is weak or missing, with no clear explanation of the problem's significance or relevance. (02)
b.	Objectives & Scope 10 Marks	Objectives are clear, specific, and measurable. The project scope is well-defined and realistic, with clear boundaries and achievable goals. (10)	Objectives are clear but may not be fully measurable or specific. The scope is defined but could benefit from more detail or clearer boundaries. (08)	Objectives are somewhat vague or broad, lacking specificity or measurability. The scope is defined but may be too broad or narrow, with unclear goals. (05)	Objectives are unclear, vague, or unrealistic. The scope is poorly defined, with unclear or unachievable goals, leading to potential project failure or misdirection. (02)

c.	Innovation and Originality 10 Marks	The project is highly innovative, presenting a novel solution or approach. The proposed project contains many significantly original, innovative, or creative aspect(s) (10)	The project shows good innovation, with some novel elements or approaches. Project contains some original, innovative, or creative aspect(s). (08)	The project shows some innovation, but the concept may be derivative or lack originality. Novelty in the approach or methodology is limited. (05)	Limited information Only Some objectives of the proposed work are defined. (02)
d.	Technical Feasibility 10 Marks	The project is technically feasible, with a clear, well researched plan for implementation. Required tools, technologies, and resources are well-defined and appropriate. (10)	The project is technically feasible, though some aspects may lack detail or thorough research. Tools and technologies are mostly defined and appropriate. (08)	The project is technically feasible, but with significant gaps in the plan or resources. Some tools or technologies may be inappropriate or poorly defined. (05)	The project is technically infeasible, with major gaps in the plan or required resources. Tools, technologies, and resources are unclear, inappropriate, or missing. (02)
e.	Methodology & Approach 15 Marks	The methodology is well defined, detailed, and appropriate for the project. The approach is logical, with clear steps, milestones, and deliverables aligned with objectives. (15)	The methodology is clear and appropriate, though some details or steps may be underdeveloped. The approach is logical, but milestones or deliverables could be clearer. (12)	The methodology is basic, with several gaps or unclear steps. The approach may lack logical flow, and milestones or deliverables are vaguely defined. (07)	The methodology is poorly defined or inappropriate. The approach lacks logic or coherence, with unclear or missing milestones and deliverables. (05)

f.	Expected Outcomes & Impact 20 Marks	Expected outcomes are clearly defined, realistic, and aligned with the project objectives. The potential impact is significant, with well-articulated benefits to stakeholders. (20)	Expected outcomes are clear and realistic but may lack full alignment with objectives. The potential impact is good, but benefits could be better articulated. (15)	Expected outcomes are defined but may be vague or unrealistic. The potential impact is limited, with benefits that are unclear or poorly articulated. (10)	Expected outcomes are unclear, unrealistic, or poorly defined. The potential impact is minimal or not well explained, with unclear benefits to stakeholders. (05)
g.	Project Timeline & Milestones 10 Marks	A detailed and realistic timeline is provided, with clear milestones and deadlines. The timeline aligns well with the project scope and objectives. (10)	A timeline is provided with clear milestones, though some details or deadlines may be vague. The timeline generally aligns with the project scope and objectives. (08)	A basic timeline is provided, but with significant gaps or unrealistic deadlines. Milestones may be unclear or misaligned with project objectives. (05)	The timeline is poorly defined, unrealistic, or missing critical milestones. Deadlines may be unclear, unrealistic, or misaligned with project objectives. (02)
h.	Documentation Quality 15 Marks	The proposal is exceptionally well-organized, professional, and free of grammatical errors. Formatting is consistent, with a clear and logical flow throughout the document. (15)	The proposal is well-organized and mostly free of grammatical errors. Formatting is consistent, with minor issues in flow or presentation. (12)	The proposal has some organizational issues or grammatical errors. Formatting may be inconsistent, affecting overall readability and presentation. (07)	The proposal is poorly organized, with numerous grammatical errors. Formatting is inconsistent, leading to difficulty in understanding and following the content. (05)

CAPSTONE PROJECT IN DATA SCIENCE II FINAL PRESENTATION EVALUATION RUBRICS

LEVELS OF ACHIEVEMENTS (Total 100 marks)					
		Excellent	Good	Average	Poor
a.	Refinement and Implementation of Solution 10 marks	Demonstrates clear, comprehensive refinement of the initial solution with well-documented iterations and improvements. All components are efficiently implemented with high technical depth and correctness. (10)	Shows good refinement of initial solution with moderate improvements. Implementation is mostly complete and technically sound. (07)	Minimal refinement of initial idea; implementation shows basic improvements but with limitations or technical flaws. (05)	Very limited or no refinement; poor implementation quality with major gaps. (02)
b.	Justification of Technology Choices and Methodologies 10 marks	Clearly explains and justifies all choices of technologies and methods based on data and results; alternatives are critically evaluated. (10)	Provides reasonable justification for technologies/methods used, with some comparison to alternatives. (07)	Justification is present but lacks depth or is based on assumptions; no clear comparison. (05)	Inadequate or no justification of technology choices or methodology. (02)
c.	Creativity and	Demonstrates	Shows originality and	Shows some	Lacks originality and

	Innovation 15 marks	exceptional originality and innovation in addressing the problem, offering new and creative solutions. Displays a high level of creativity in the project's design or implementation with unique approaches. (15)	innovation in addressing the problem, though it may not be highly distinctive. Demonstrates creativity in design or implementation but with some conventional elements. (10)	originality and innovation, but may be somewhat predictable or conventional. Displays basic creativity in design or implementation, but lacks unique or standout elements. (05)	innovation, offering conventional solutions that do not address the problem uniquely. Minimal or no creativity in the project's design or implementation. Most of the components are plagiarized. (03)
d.	Quality of Final Outcome/Product 20 marks	Final product/outcome is complete, highly functional, technically strong, and well-validated. Reflects excellent understanding and effort. Student has a clear understanding of what has been done and has completed all the tasks. Student is fully focused on the work and always trying to give a better outcome. (20)	Outcome is functional and meets most expectations; technically adequate with minor issues. Student has a clear understanding of what is done but looks like outside help was obtained. Students are trying to focus on the work and to give a better outcome. (15)	Outcome works partially but lacks completeness or technical depth. Student somewhat understands what was done, but looks like that mostly outside help was obtained. Students are not fully focused on the work. But trying to give some better outcome. (10)	Outcome is incomplete, non-functional, or poorly developed. Student has done very little work and obtained help from others. Quality of the work is insufficient. (05)

e.	Depth of Data Analysis and Evaluation 20 Marks	Provides in-depth analysis using advanced techniques, with strong statistical validity and interpretation. Results are critically evaluate (20)	Analysis is complete and appropriate, with reasonable interpretation and some critical evaluation. (15)	Analysis is done but may miss depth or interpretation; evaluation is surface-level. (10)	Incomplete or flawed analysis; interpretation lacks logic or is missing. (05)
f.	Maintaining Professionalism and Quality of the presentation 15 marks	Student has performed in an excellent way with; • Clear sound • Better proficiency in language • Proper body gestures • Good tonality • Good eye contacts • Proper time management • Wearing suitable attires (15)	Student has performed well with; • Good sound • Good language • Proper body gestures • Good tonality • Good eye contacts • Proper time management • Wearing suitable attires (10)	Student has performed in an averagely with; • Less sound • Good language • Body gestures • Tonality • Eye contacts • Time management Suitable attires (05)	Student performances are not adequate with; • Poor sound • Proficiency in language • Poor body gestures • Poor tonality • Poor eye contacts • Poor time management Not wearing suitable attires (03)

g.	Handling the Q & A 10 marks	Student has listened to question carefully, respond directly to the question asked and answered the question without defensiveness (10)	Student has listened to question, respond directly to the question asked and answered the question without defensiveness (06)	Students have listened to the question partially, not respond directly to the question asked and answered the question without defensiveness on average. (04)	Student has not listened to question carefully, not respond directly to the question asked and answered the question with defensiveness (02)
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CAPSTONE PROJECT IN DATA SCIENCE I FINAL REPORT EVALUATION RUBRICS

LEVELS OF ACHIEVEMENTS					
		Excellent	Good	Average	Poor
a.	Introduction & Background 15 Marks	The introduction provides a clear, concise overview of the project, including a well-defined problem statement and context. Background research is thorough and relevant, clearly establishing the need for the project. (15)	The introduction is clear but may lack some detail or depth. The background research is adequate but may miss some key aspects or connections to the problem. Problem statement is clearly described, but some key details are omitted. (10)	The introduction is somewhat vague or lacks clarity. Background research is present but may be superficial or only partially relevant to the problem. (05)	The introduction is unclear or poorly written, with little to no background research. The problem statement is vague or missing, making the project's purpose unclear. (03)
b.	Project Objectives / Scope /Motivation 15 Marks	The objectives are clearly defined, specific, and measurable. The scope is well-defined and appropriate, with clear boundaries and alignment with the project's goals. The motivation for pursuing the project and its relevance are clearly and persuasively established by relating the project to related work.	The objectives are clear but may lack full specificity or measurability. The scope is defined but may be slightly too broad or narrow in relation to the project's goals. The motivation for pursuing the project is somewhat clear, but no support is provided. (10)	The objectives are somewhat vague or overly broad. The scope is unclear or not fully aligned with the project's goals, leading to potential confusion. The motivation of the project is not clear or described. (05)	The objectives are unclear, vague, or unrealistic. The scope is poorly defined, with significant misalignment to the project's goals, leading to project misdirection. (02)

		(15)			
c.	Design & Methodology 30 Marks	The report provides a detailed, logical, and well-organized description of the design and methodology. The approach is appropriate for achieving the project's objectives, with clear justification for design choices. (30)	The design and methodology are well-described but may lack some detail or depth. The approach is generally appropriate, though some aspects could be better justified. (20)	The design and methodology are described but lack clarity or thoroughness. The approach may be somewhat inappropriate or poorly justified for the project's objectives. (10)	The design and methodology are poorly described or missing. The approach is inappropriate or unjustified, leading to potential failure in achieving the project's objectives. (05)
d.	Implementation, Results, Conclusion, and Discussion 20 Marks	The implementation is thoroughly documented, with clear explanations of the steps taken. Results are well-presented, with appropriate use of tables, graphs, and analysis. Outcomes are clearly linked to the objectives. Project work is well summarized and concluded. Future extensions in the project are well	The implementation is well documented, but some details may be unclear or missing. Results are presented adequately, but analysis may be lacking in depth. The project work summary and conclusion are not very appropriate. Future extensions in the project are specified. (15)	The implementation is documented, but with significant gaps or unclear explanations. Results are presented, but may be poorly organized or lack thorough analysis. The project work summary and conclusion are not very appropriate. Future extensions in the project are not specified. (10)	The implementation is poorly documented or incomplete. Results are unclear, poorly presented, or missing. Analysis is minimal or non-existent, failing to link outcomes to objectives. Project work is not summarized and concluded. Future extensions in the project are not specified.

		specified. (20)			(05)
e.	References & Citations 10 Marks	The report includes a wide range of high-quality references that are relevant, up-to-date, and properly cited. The citations follow a consistent and appropriate style, adding credibility to the report. (10)	The report includes several good references, though some may be slightly outdated or less relevant. Citations are mostly correct, with minor inconsistencies in style. (07)	The report includes a few references, but they may be of questionable quality or relevance. Citations are inconsistent or improperly formatted. (05)	The report includes minimal or no references, or the references used are of poor quality or irrelevant. Citations are missing, incorrect, or inconsistent in style. (02)
f.	Report Formatting and Quality of Writing 10 Marks	Reports are formatted exactly as mentioned in the guidelines. Complete explanation of the key concepts and strong description of the technical requirements of the project Outstanding, comprehensive and clear report. (10)	Reports are formatted as mentioned in the guidelines. Complete explanation of the key concepts but insufficient description of the technical requirements of the project Effective report using academic language. (07)	Reports are formatted as mentioned in the guidelines at a some certain level Incomplete explanation of the key concepts and insufficient description of the technical requirements of the project. Acceptable report structure, no missing parts, clarity of language	Reports are poorly formatted. Report is unbalanced or unclear, or it is difficult to follow ideas. Report is unreadable as an English report Inappropriate explanation of the key concepts and poor description of the technical requirements of the project. Major sections are missing.

				(05)	(02)
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